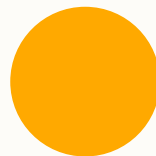




Software Development

Reese Hatfield



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Careers

- We've touched on a LOT of different topics in CS throughout the scope of this course
- From binary computing to quantum computing
- What can you actually do with it tho?





Careers

- Right now especially, people want to know what you can do with a computer science background
- Rise of AI
- Where can you go with that career?





Careers

- Tackle this in two pieces
 - Software fields
 - Non-software fields
- Software fields are what everybody wants
 - And what they are all worried about





Careers

- Software Developer
- Is the primary career in computer science
- Very broad
- We'll talk about what that tends to look like today





Careers

- Software Development:
- The people who code and turn that code into usable software
- “Coding/Programming” is the primary responsibility of developer
- But how does this work in practice?





Careers

- In class, we did small, toy projects
- **Considerations:**
 - Just one person
 - All the code in just one file
 - No large amount of data
 - Definite requirements





Careers

- Software developers often work in teams
- How can multiple people code a single project?
- Send the file back and forth?
- Multiple files?





Careers

- Developers need a way of all working on code at the same time
- Developers ALSO need a way to tracking how software changes over time
- **Version Control Systems**





Careers

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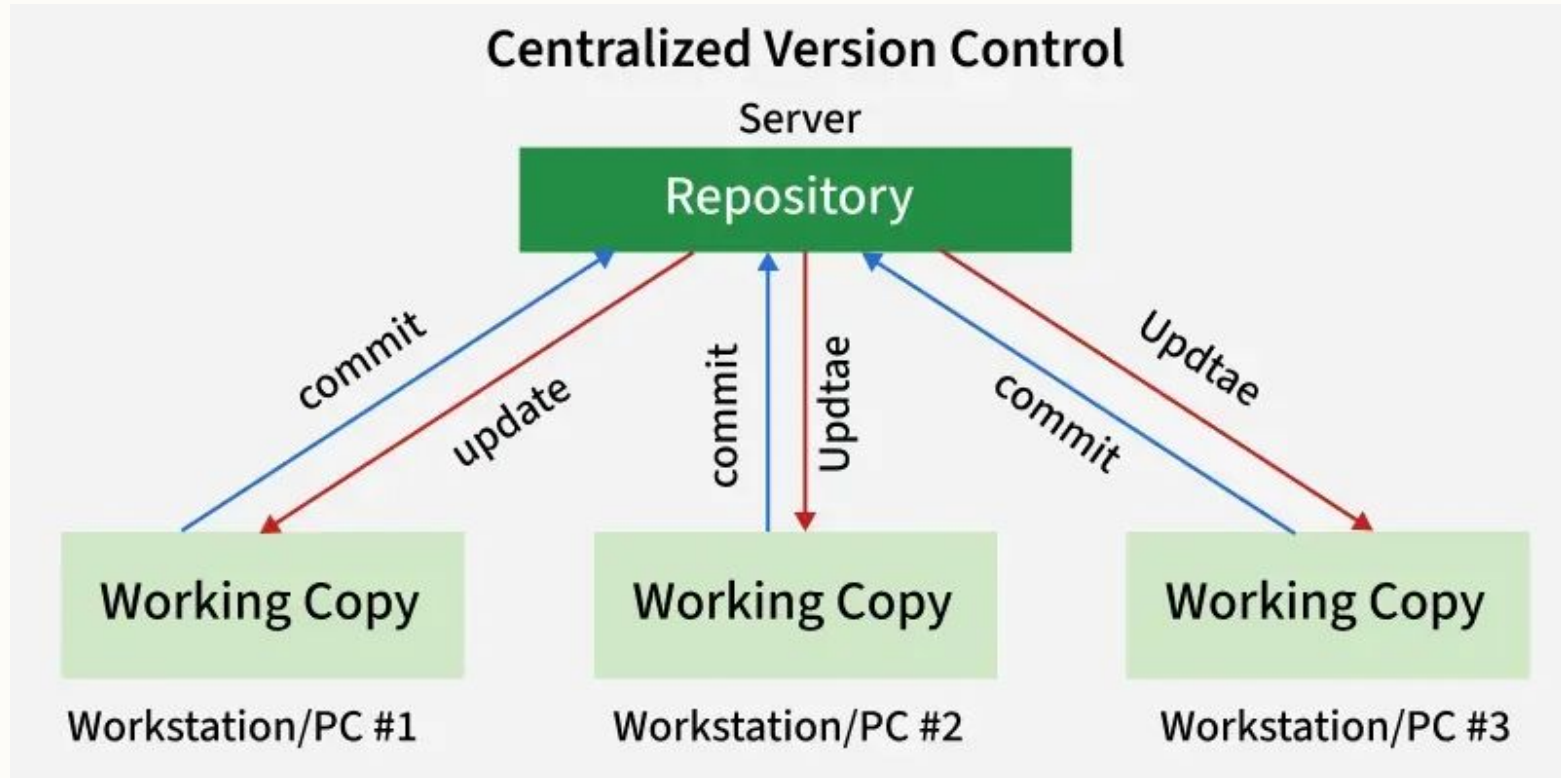


Careers

- **Version Control Systems**
- Allow many developers to work on the same piece of code at the same time
- This might feel similar to Google Docs
- They use a **VCS** under the hood



Careers





Careers

- Codebase $\Rightarrow$ large repository of code needed to build the software
- A **commit** represents some change to the code
- All developers update their code when someone else commits
- Mostly automated*





Careers

- Changes like this, are usually **committed** to a canonical copy of the software
- Often, this copy doesn't even exist on someones system
- All develops **pull** their changes onto a local copy





Careers

- **Git** is the most popular version control
 - This can't be understated
- **VCS** is synonymous with **Git**
- **Linus Torvalds** creation





Careers

- Git works as a directed graph implemented as code, under the hood
- A **commit** is a node in the graph
- You can rebuild the code by looking at ALL of the commits it took to get there



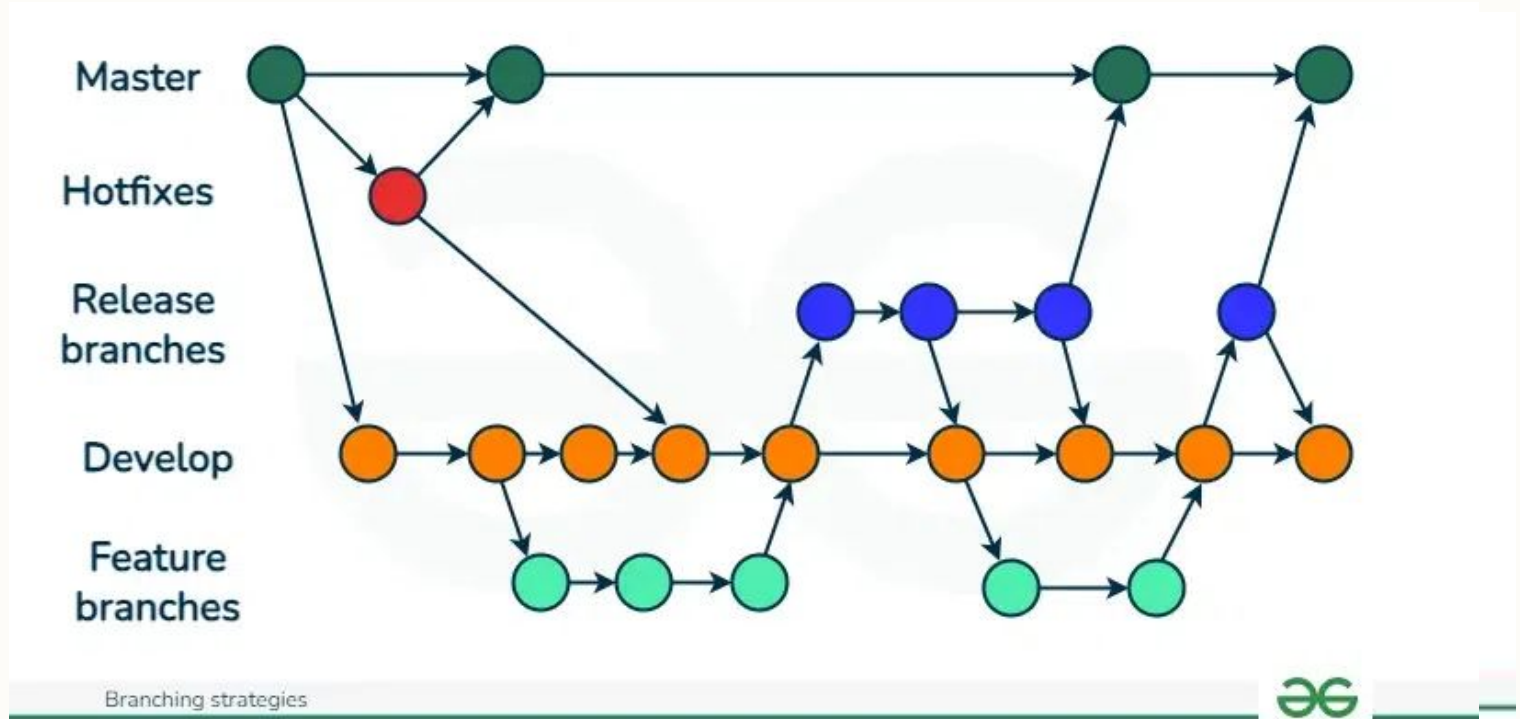


Careers

- Developers create **branches** to diverge from a particular commit
- Its almost like a tree structure*
- Branch is just another path of commits
- Need to be re-merged back into the original point



Careers





Careers

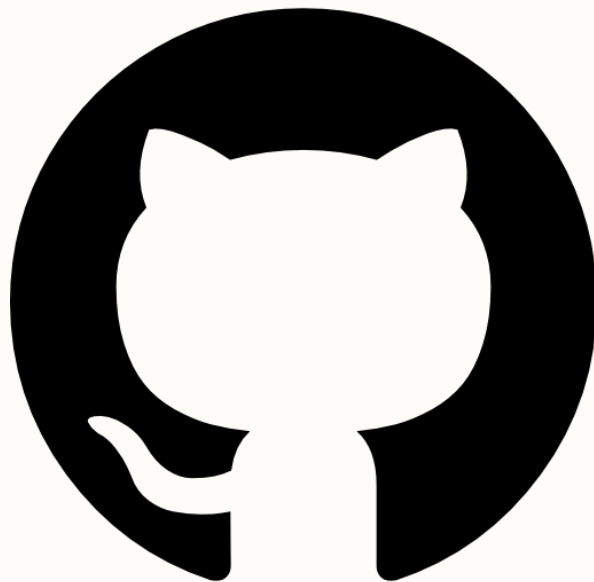
- Git is a **centralized** version control system
- Remember, one canon copy of the codebase exists somewhere
- Someone has to do all of that hosting
- Who is responsible for that?





Careers

- **Github** is the primary git repository hosting platform
- We use git to track our files for this class
- Let's look at some repos





Careers

- Developers need a way to track things that need fixed
- This is sometimes done in external software, think Jira
- However, this is VERY often integrated directly into the git host platform





Careers

- Github has a system in place called *issues*
- Create an Issue
- Build a backlog of issues
- Developers can be assigned issues
- Code them, close them





Careers

- Public codebases on git hosts like Github
- Are the fundamental backbone of open source software development
- OSS pays no money
- For the betterment of society





Careers

- Many apps can really be broken down into three pieces developers can work on
- Frontend: What the user sees
 - Website, UI, gameplay, etc
- Backend: What the user doesn't see
 - Servers, database, etc
- Database: Where the data get's stored





Careers

- Many developers have some kind of preference
- Web Developers $\Rightarrow$ Frontend, etc.
-
- What are the roles of each kind of developer?





Careers

- **Frontend Developers**
 - Often a web target
- What Javascript framework to use?
- CSS math
- Dynamic page loading
- Fast load times
- Memory balancing





Careers

- **Backend developers:**
 - Often target a linux server
- How to interact with the database?
- How to set up a good JSON API?
- How to handle hundreds of query?



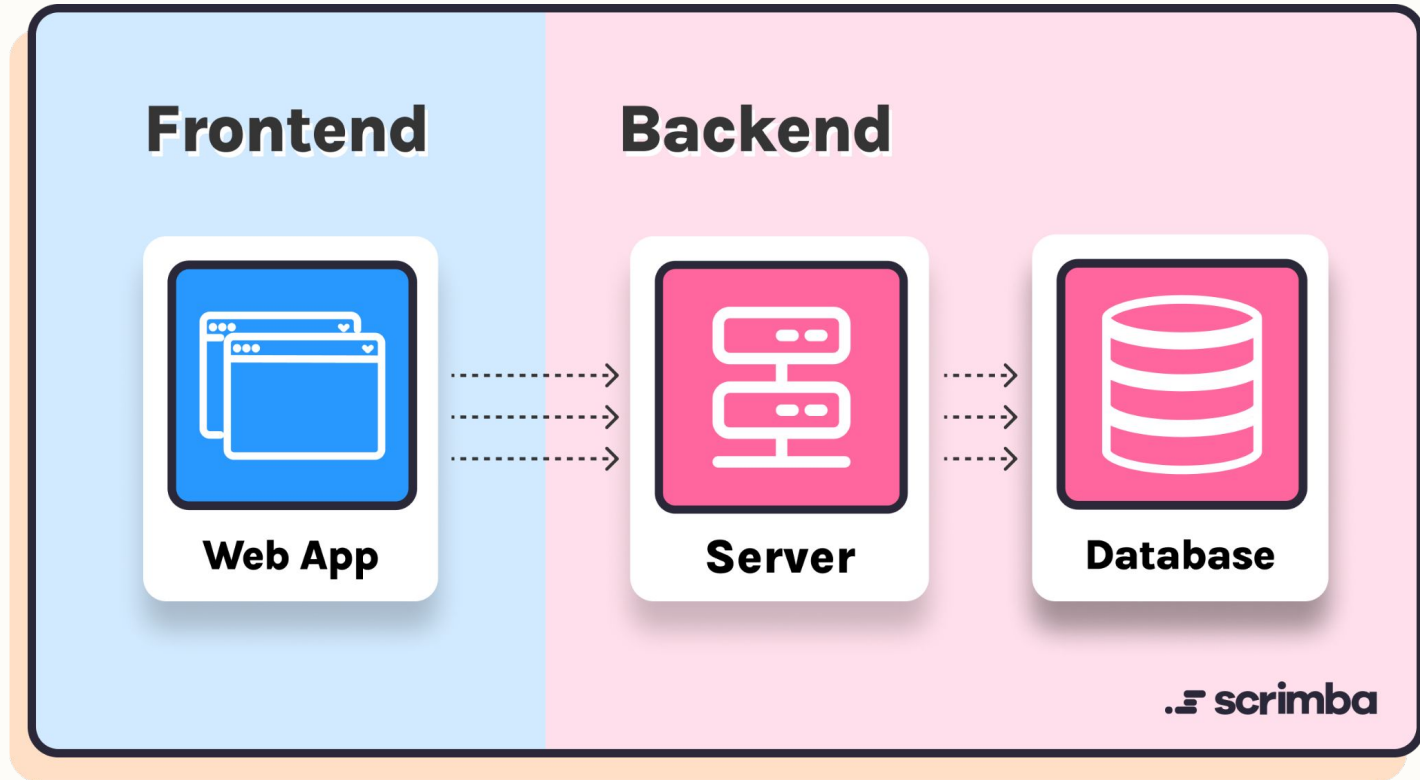


Careers

- **Database Engineers**
 - Designing big data storage
- Database normalization
- Scalability
- Caching
- Relational vs. Key-Value



Careers





Careers

- **Three Tiered Application**
- Framework for pretty much all modern applications
- It works, regardless of your tech stack
 - Set of frameworks and technologies you use for your project
- Helps delegate work

